UP Board 2026 Class 12 Biology 348 (EM) Question Paper

Time Allowed :3 Hours 15 Minutes	**Maximum Marks :**70	**Total Questions :**09

General Instructions

Read the following instructions very carefully and strictly follow them:

1. First 15 minutes are allotted for the candidates to read the question paper.
2. All questions are compulsory.
3. Illustrate your answers with labelled diagram wherever necessary.
4. Marks allotted to each question are mentioned against it.

Multiple Choice Type Questions

1. (a) Pollination occurs in *Vallisneria* –

(A) By air
(B) By insects
(C) By water
(D) None of these

(b) Who proposed the theory of 'Natural Selection'?

(A) Mendel
(B) Lamarck
(C) Darwin
(D) Debrics

(c) Enzyme that cuts DNA at a specific location is –

(A) Endonuclease
(B) Exonuclease
(C) Polymerase
(D) All of these

(d) Energy flow in the ecosystem is –

(A) Unidirectional
(B) Bidirectional
(C) Multidirectional
(D) All of these

Very Short Answer Type Questions

2. (a) Which hormone is secreted from the corpus luteum?

(b) What is the full name of IVF?

(c) What is the termination codon on mRNA?

(d) Give one example of sex linked recessive disorder.

(e) Which ecological pyramid is always upright?

Short Answer Type Questions – I

3. (a) Write a note on Darwin's Finches.

(b) Define the word 'Health'. Explain the benefits of good health.

(c) Differentiate between m-RNA and t-RNA.

(d) What is the difference between food chain and food web?

(e) Comment upon Biodiversity hot-spots.

Short Answer Type Questions – II

4. (a) What is the difference between microsporangium and megasporangium?

(b) Draw a labelled diagram of the sectional view of seminiferous tubules.

(c) Explain transcription unit.

(d) Explain the causes, prevention and treatment of Malaria disease.

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5. (a) What is mutation? Explain with an example.

(b) What do you understand by biocontrol? Explain about their benefits.

(c) Explain gel-electrophoresis.

(d) Explain the different types of biodiversity.

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6. (a) What is Allergy? Explain with examples.

(b) Write a short note on Pest Resistant Plant.

(c) Write a short note on infectious diseases.

(d) What do you understand by Gel Electrophoresis? Explain it.

Long Answer Type Questions

7. What is microsporogenesis? Describe microsporogenesis in angiospermic plant

with diagram.

OR

Describe the process of spermatogenesis with a diagram.

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8. What do you understand by DNA fingerprinting technique? Explain its process.

OR

Write short notes on the following –
(i) Translation
(ii) Hardy–Weinberg Principle.

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9. Describe the different types of ecological pyramids with a diagram.

OR

What is decomposition? Describe the decomposition cycle in a terrestrial ecosystem with a diagram.
